

Some Software Tools Updated in the Last One Year

Category	Name	Licence	Description	Latest features
Analogue simulator	AIM-Spice 5.9	Free	Automatic Integrated Circuit Modeling Spice (AIM-Spice) uses SPICE format for describing analogue circuits and has point-and-click type controls	• Supports BSIM4v8.0 • Supports ND TNET model version 1.6 • Versions for Android and iOS released
cali	VariCAD 2015-2.0	Licensed	Tools for 3D modelling and 2D drawing-cum-editing, and support for geometrical constraints; tools for shells, pipelines, sheet metal unbending and crash tests, and assembly support	• Graphic based on OpenGL 4.0 or 4.3, making rendering 2.5x to 10x faster • File opening and creation of graphic data is approximately 2x faster • New version automatically turns on anti-aliasing, making 3D images look significantly better
Data logger and analyser	CAPture VDAS (CAPture Data Logger)	Licensed	Acquires, displays, logs physical parameters from connected sensors; can also store and analyse data	
Electronic CAD	LASI 7.0.88	Free	LAYout System for Individuals (LASI), originally intended for integrated circuits, can be used for ICs, MEMS, other technologies, discrete devices, schematics and project documentation drawings	• Edulib cell library and basic tutorial have been revised to include Top.gds and Opamp.gds file examples of GDS conversion • Even byte strings are required by GDS format • Imager tutorial now contains Spice CIR file Imager_a_11.cir compiled from cell Imager_a_11 by LasiCkt utility • Maximum text size has been increased by a factor of 10
Electronics design environment	Altium Designer 15	Licensed	Integrated environment that includes tools for schematic capture, PCB design, FPGA design, embedded software development and manufacturing file support	• xSignals to accurately plan and route high-speed designs • Solder mask-expansion enhancements by which top and bottom layers can be configured individually • Separate 2D- and 3D-view orientations • Supports collaboration between electrical and mechanical computer-aided design (ECAD and MCAD) domains with incremental design exchange (IDX) format
Embedded design	Kactus2 2.8.0	Free	Metadata based software for sketching, packing, integrating and generating both hardware and software for embedded products at several hierarchy levels	• Feature to import parameters from included files • Editors for IP-XACT elements remapState and memoryRemap • Support for expressions in memory map elements • Support for parameter references in generated header files • Feature to save new versions of design configurations and designs • Support for module parameters to hierarchical views
FPGA CAD	RapidSmith 0.5.2	Free	Easy-to-use platform to work on modern Xilinx FPGAs. Offers a compact and fast device database with several APIs to enable different functions for Xilinx devices	• Preliminary support for Artix 7 and Zynq devices • Preliminary support for the inout pin type found in Zynq device/XDL files
HDL	MyHDL 0.8.1	Free	Python based hardware description and verification language. Designs can be converted to Verilog or VHDL automatically and implemented using a standard tool flow	• Elegant modelling of wrap-around behaviour with modular bit-vector types • A new decorator for error-free reset behaviour coding • Adds conversion of top-level class methods
IC design	Symica FE / Symica Toolkit 2.3.0	Dual	Integrates schematic editor, SPICE simulator, waveform viewer and analyser and other additional features, such as library manager, symbol editor, hierarchy editor and input and output translators; paid version includes support for mixed-signal designs and unlimited number of devices, threads and Verilog-A model parameters	• Creates different netlists for different tools like SPICE, LVS and others • Probing currents for complex transistor models represented as sub-circuits • Shares project among multiple designers
PCB design	ZenitPCB 1.8.1	Free	Easy-to-use program that lets you quickly create professional-quality PCBs	
Programming language and numerical computing environment	Octave 4.0	Free	Free and close alternative to MATLAB for writing numerical solution of linear and nonlinear problems, and performing other numerical experiments	• Equipped with graphical user interface (GUI) along with traditional command line interface (CLI) • Support for classdef object-oriented programming • Better compatibility with MATLAB